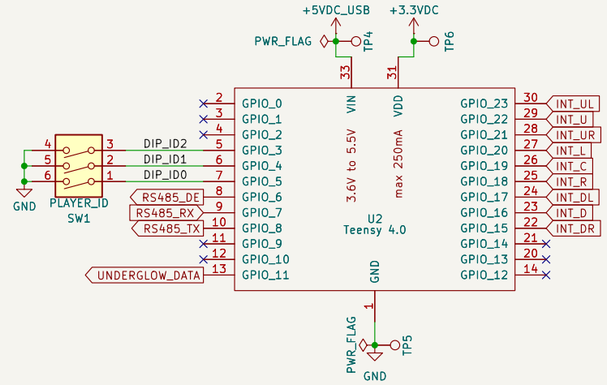
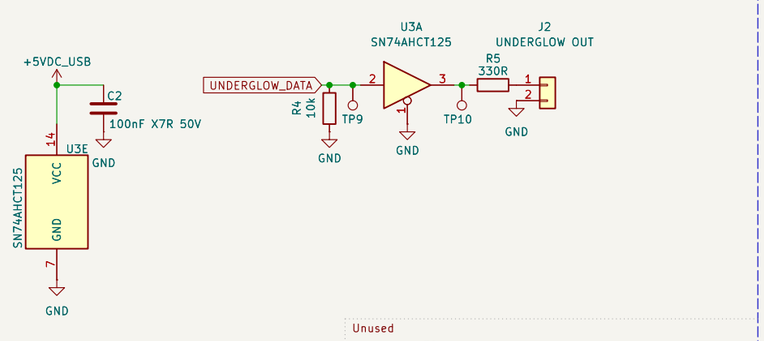
Microcontroller

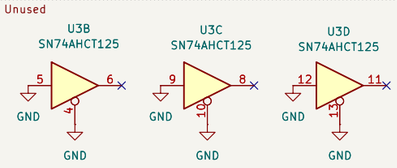


Underglow LED Logic Shifter & Ground Hookup

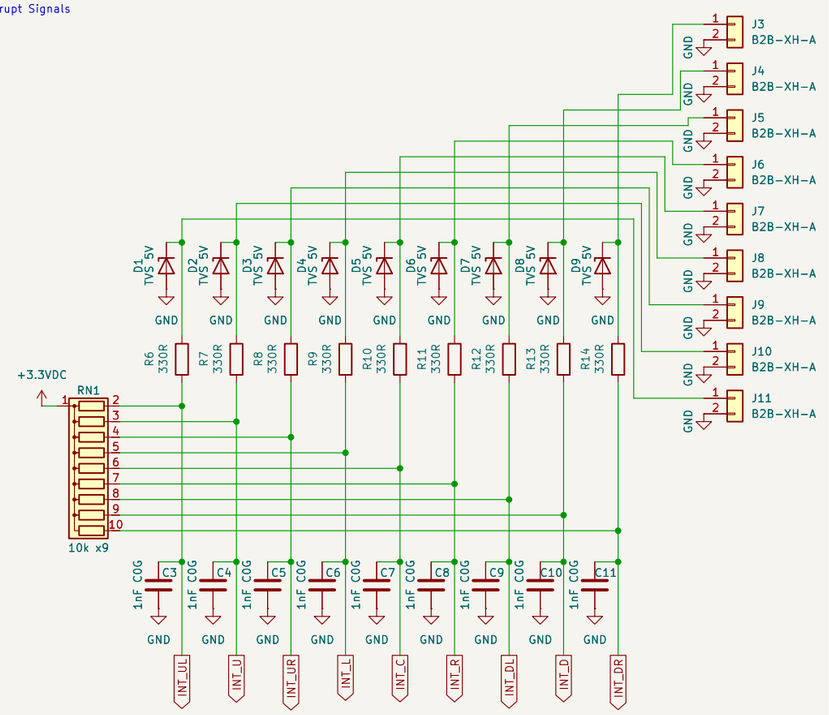
Overall Ground connection is added here to J4, as 2P terminal blocks are easy to find and source. We needed a way to connect the ground to the rest of the system anyway.



Unused

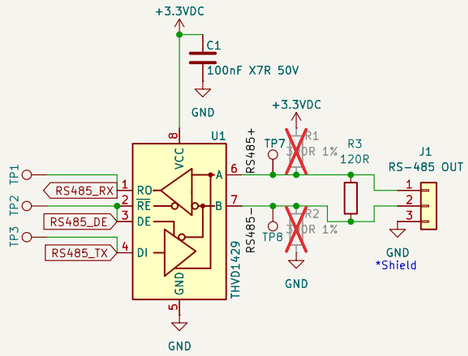


Panel Interrupt Signals



RS-485 Receive & Transmit

RS-485 bias (R4/R5): fitted as DNP footprints, insurance only.
THVD1429 has integrated open/short/idle failsafe, so the idle bus is already defined and no external bias is normally needed.
Bias belongs at ONE point on the bus — the master is that point, so if the bench ever disagrees, populate here and nowhere else.
390R pair = approx. 236mV across the 60R terminated bus at 3.3V.



Mounting

USB-powered; NO 12V on this board. See docs/MASTER_SCHEMATIC_PLAN.md
Teensy 4.0 master: 9x INT in, RS-485 out, underglow data out, player-ID DIP

Sheet: /	
File: master-pcb.kicad_sch	
Title: KrakenPad Master PCB	
Size: A3	Date: 2026-07-18
KiCad E.D.A. 10.0.4	Id: 1/1

Rev: 0.1